TutoConnect

Software Testing Document

Sprint 1 — Authentication & Search

Section 7, Group 4 | Principles of Software Engineering

Document Type	Software Testing Document — Sprint 1
Testing Types	User Acceptance Testing (UAT) · System Testing · Integration Testing
Testing Approach	Prototype mockup inspection. Testers view each screen and verify that the visible UI elements match what is expected. No live backend or data entry is performed.
Mockup Screens	Sign In, Registration (role selector), Student Registration, Tutor Registration, Recover Password, OTP Entry, Update Password, Search Page, Filtered Results, Filter Popup, Language Dropdown, Subjects Dropdown, Study Level Dropdown, Side Menu
Roles Evaluated	Student, Tutor
Version	1.0

1. Introduction

1.1 Purpose

This document records the testing conducted on the TutoConnect Sprint 1 prototype mockups. The prototype consists of static screen designs showing the intended UI for each Sprint 1 feature. All testing is performed by visually inspecting each screen and checking whether the correct elements are present. No data is entered, no backend calls are made, and no real accounts exist in the prototype.

1.2 How Tests Are Conducted

The tester opens the relevant mockup screen and observes what is visible. Each test case states: (1) which screen to look at, (2) what elements should be visible based on the user story or design requirement, and (3) what was actually observed on the mockup. A Pass means all expected elements are present. A Fail means one or more required elements are missing from the mockup.

1.3 Mockup Screens Referenced

Sign In page · Registration page (role selector) · Student Registration page · Tutor Registration page · Recover Password page · OTP Entry page · Update Password page · Search page · Filtered Results page · Filter Popup · Language Dropdown · Subjects Dropdown · Study Level Dropdown · Side Navigation Menu

2. User Acceptance Testing (UAT) — Sprint 1

Each test references a specific mockup screen. The tester views that screen and checks whether the listed elements are visible as described.

Test ID	User Story	Mockup Screen	What Was Checked	What Is Visible on the Mockup	Status
UAT-S1-01	US-01: Role selection on registration	Registration Page (role selector state)	The screen must show a role selector so the user can choose Student or Tutor before seeing role-specific fields.	Registration page shows a shared form with 'I am a:' followed by Student and Tutor radio buttons. Student is pre-selected (filled blue dot). Shared fields visible: Full Name, Email, Username, Password, Confirm Password, Upload Student ID, Institution.	**Pass**
UAT-S1-02	US-01 & US-02: Student registration — full field set	Student Registration Page (Student selected, fully expanded)	With Student selected, the form must show all student-specific fields: Learning Languages dropdown, Subjects of Interest dropdown, Enter Level of Studies dropdown, and a Register button, in addition to the shared User fields.	Student Registration page shows all fields in order: Full Name, Email, Username, Password, Confirm Password, Upload Student ID (with upload icon), Institution, Learning Languages (dropdown with chevron), Subjects of Interest (dropdown with chevron), Enter Level of Studies (dropdown with chevron), green Register button.	**Pass**
UAT-S1-03	US-03: Subject and language dropdowns show selectable options	Subjects Dropdown screen, Language Dropdown screen	The Subjects dropdown must list subject options. The Language dropdown must list language options. Both must show a 'More...' option.	Subjects Dropdown: Mathematics, Physics, Chess, History, More... Language Dropdown: English, Spanish, Chinese, French, More... Both screens present with correct options.	**Pass**
UAT-S1-04	US-02: Level of studies dropdown shows academic levels	Study Level Dropdown screen	The level of studies dropdown must list academic levels.	Study Level Dropdown shows: Degree, Masters, Phd, Diploma, More... All expected options present.	**Pass**

UAT-S1-05	US-04: Student ID file upload control is present	Student Registration Page	An Upload Student ID control with an upload icon must be visible on the student registration form.	'Upload Student ID' field with upload icon (↓) is visible on the Student Registration page.	**Pass**
UAT-S1-06	US-05 & US-06: Tutor registration — full field set	Tutor Registration Page (Tutor selected)	With Tutor selected, the form must show: Teaching Languages dropdown, Subjects Taught dropdown, Upload Qualifications file control, and a Register button, in addition to the shared User fields.	Tutor Registration page shows all shared User fields plus: Teaching Languages dropdown, Subjects Taught dropdown, Upload Qualifications (with upload icon), green Register button. Tutor radio button is filled blue (selected).	**Pass**
UAT-S1-07	US-07: Sign In screen elements	Sign In Page	Sign in screen must show: Username field, Password field, Remember Me checkbox, Login button, Register link, and Forgot Password link.	Sign In page shows: 'Sign in' heading, Username field (person icon), Password field (lock icon), 'Remember me' checkbox (ticked, filled blue), green Login button with arrow icon, 'Don't have an account? Register', and 'Forgot your password?' text.	**Pass**
UAT-S1-08	US-07: Error state for wrong credentials	Sign In Page	An error message area or placeholder must be visible on the sign in screen to represent the invalid credentials feedback state.	No error message area or error placeholder is visible on the Sign In mockup. The invalid credentials state is not represented.	**Fail**
UAT-S1-09	US-08: Remember Me checkbox	Sign In Page	A 'Remember Me' checkbox must be visible on the sign in screen.	'Remember me' checkbox is visible and shown as ticked (filled blue) on the Sign In page.	**Pass**
UAT-S1-10	US-09: Forgot Password link on Sign In	Sign In Page	A 'Forgot your password?' link must be visible on the sign in screen.	'Forgot your password?' is visible below the Login button on the Sign In page.	**Pass**

UAT-S1-11	US-09: Recovery screen 1 — enter email	Recover Password Page	A screen with an email input field and a Send OTP button must exist as the first password recovery step.	Recover Password page: 'Recover Password' heading, Enter Email field (email icon), green Send OTP button, 'Already have an account? Signin' link.	**Pass**
UAT-S1-12	US-09: Recovery screen 2 — enter OTP	OTP Entry Page	A screen with an OTP input field and a Reset Password button must exist as the second recovery step.	OTP page: 'Reset Password' heading, Enter OTP field (icon), green Reset Password button, 'Already have an account? Sign in' link.	**Pass**
UAT-S1-13	US-09: Recovery screen 3 — set new password	Update Password Page	A screen with New Password and Confirm Password fields must exist as the third recovery step.	Update Password page: 'Update Password' heading, New Password field (lock icon), Confirm Password field (lock icon), green Login button.	**Pass**
UAT-S1-14	US-10: Logout in navigation	Side Menu	A Logout option must be visible in the navigation menu for logged-in users.	Side Menu shows: TutoConnect header, Dashboard, Search, and 'Log out' at the bottom with an exit icon.	**Pass**
UAT-S1-15	US-11: Search bar on Search page	Search Page	A search bar with a filter icon and a search submit icon must be visible on the search page.	Search page: search bar with blue filter funnel icon on the left and magnifying glass icon on the right. 'Search' placeholder text visible.	**Pass**
UAT-S1-16	US-11: Tutor result cards on Search page	Search Page (default results state)	Tutor cards must be visible showing: name, subject, price per hour, star rating, and short bio.	Search page shows 6 tutor cards in a 2-column grid. Each card: avatar icon, tutor name, subject (e.g. 'Principle of software engineering'), price (e.g. '250rm/hour' in blue), 5-star rating in yellow, short bio text.	**Pass**
UAT-S1-17	US-12: Filter popup controls	Filter Popup	Filter popup must show: Subject dropdown, Language dropdown, Min Price field, Max Price field,	Filter Popup: Subject dropdown, Language dropdown, Min Price ($ field), Max Price ($ field), Stars (with star icon), green Apply button, grey	**Pass**

			Stars control, Apply button, Reset button.	Reset button, X close button top-right.	
UAT-S1-18	US-12: Filtered results shown after filter applied	Filtered Results Page	Results page must show tutor cards matching the applied filter with the search term visible in the search bar.	Filtered Results: search bar shows 'Chess' with active filter icon. 6 cards visible, all labelled 'Chess instructor' with varying prices (200–1000rm/hour) and 5-star ratings. Tutors: Magnus Carlsen, Daniel Naroditsky, Levon Aronian, Kevin Bordi, Judit Polgar, Baadur Jobava.	**Pass**

3. System Testing — Sprint 1

Derived from Sprint 1 Sequence Diagrams (SD1–SD4) and ERD. Each test verifies that the mockup screens together represent the full system flow — checking that all steps in a flow are visually covered.

Test ID	System Flow (SD / ERD Ref.)	Screens Inspected	What Was Checked	Observation	Status
ST-S1-01	SD1 — Registration: USER + STUDENT records. ERD: USER fields (fullName, email, username, passwordHash, institution) + STUDENT fields (subjects, languages, level, studentIDFile)	Student Registration Page	All ERD USER and STUDENT entity fields must have a corresponding visible input on the student registration screen.	Full Name ✓, Email ✓, Username ✓, Password ✓, Confirm Password ✓, Institution ✓, Learning Languages ✓, Subjects of Interest ✓, Level of Studies ✓, Upload Student ID ✓. All ERD fields covered.	**Pass**
ST-S1-02	SD1 — Registration: USER + TUTOR records. ERD: TUTOR fields (teachingLanguages, subjectsTaught, qualificationsFile)	Tutor Registration Page	All ERD TUTOR entity fields must have a corresponding visible input on the tutor registration screen.	Teaching Languages ✓, Subjects Taught ✓, Upload Qualifications ✓, plus all shared USER fields ✓. All ERD TUTOR fields covered.	**Pass**
ST-S1-03	SD1 — Role routing: system determines whether to create STUDENT or TUTOR record based on role selection	Registration Page (role selector)	The role selector (Student / Tutor radio buttons) must be present to represent the branching point in SD1.	'I am a: Student / Tutor' radio buttons visible on Registration page. Both options present. Student shown as default selection.	**Pass**
ST-S1-04	SD2 — Login success path: credentials → session token → dashboard redirect	Sign In Page	Username field, masked password field, and Login button must all be present, representing	Username field ✓ (person icon), Password field ✓ (masked, lock icon),	**Pass**

			the credential submission step in SD2.	Login button ✓ (green). All three elements present.	
ST-S1-05	SD2 — Login failure path: system returns error message to client	Sign In Page	An error state or error message placeholder must be present on the sign in screen for the login failure path in SD2.	No error message area or placeholder visible on the Sign In mockup. The failure path in SD2 is not visually represented.	**Fail**
ST-S1-06	SD3 — Password recovery: 3-step flow. ERD: USER.resetOTP, USER.otpExpiry, USER.passwordHash	Recover Password Page, OTP Entry Page, Update Password Page	All three recovery screens must exist with correct fields: (1) email + Send OTP, (2) OTP input + Reset Password, (3) New Password + Confirm Password + submit.	Screen 1: Enter Email + Send OTP button ✓. Screen 2: Enter OTP + Reset Password button ✓. Screen 3: New Password + Confirm Password + Login button ✓. All three screens present.	**Pass**
ST-S1-07	SD2 — Remember Me: USER.rememberMeToken	Sign In Page	Remember Me checkbox must be on the Sign In screen, representing the remember-me token path in SD2.	'Remember me' checkbox visible and shown as ticked on the Sign In page.	**Pass**
ST-S1-08	SD2 — Logout: session terminated. ERD: USER session cleared	Side Menu	A Logout action must be present in the navigation, representing the logout sequence in SD2.	'Log out' with exit icon visible at the bottom of the Side Menu.	**Pass**
ST-S1-09	SD4 — Search: student submits keyword → TUTOR+SUBJECT queried → results returned	Search Page	Search input field, submit icon, and tutor card grid must be present. Each card must reflect TUTOR ERD	Search input ✓, magnifying glass icon ✓, tutor card grid ✓. Each card: name ✓, subject ✓,	**Pass**

			fields: name, subject, rating, hourlyRate, bio.	star rating ✓, price/hour ✓, bio ✓.	
ST-S1-10	SD4 — Filter: student applies filter conditions → filtered TUTOR results returned	Filter Popup, Filtered Results Page	Filter Popup must show all filter controls (Subject, Language, Min Price, Max Price, Stars). Filtered Results must show results matching the applied filter with the search term visible.	Filter Popup: Subject ✓, Language ✓, Min Price ✓, Max Price ✓, Stars ✓, Apply ✓, Reset ✓. Filtered Results: 'Chess' in search bar ✓, all 6 cards show Chess instructor ✓.	**Pass**
ST-S1-11	ERD — USER.passwordHash: password never stored in plaintext	Sign In Page, Registration Pages	Password fields must use masked input (lock icon, characters hidden) on all screens.	Password fields on Sign In ✓ and all registration screens ✓ use lock icon and masked appearance.	**Pass**
ST-S1-12	NFR — Navigation: Dashboard and Search accessible post-login	Side Menu	The navigation menu must show at minimum a Dashboard link and a Search link.	Side Menu: Dashboard (grid icon) ✓, Search (magnifying glass icon) ✓, Log out ✓.	**Pass**

4. Integration Testing — Sprint 1

Derived from the Sprint 1 Class Diagram. Checks whether the mockup screens reflect the class relationships between User, Student, Tutor, and SearchQuery.

Test ID	Class Relationship	Screens Inspected	What Was Checked	Observation	Status
IT-S1-01	User → Student (inheritance). Student has all User fields PLUS student-specific fields.	Student Registration Page	Student registration screen must contain both shared User fields (Full Name, Email, Username, Password, Institution) AND Student-only fields (Learning Languages, Subjects of Interest, Level of Studies, Upload Student ID).	All User fields present ✓ plus all Student-specific fields ✓. User→Student inheritance correctly reflected in the UI.	**Pass**
IT-S1-02	User → Tutor (inheritance). Tutor has all User fields PLUS tutor-specific fields.	Tutor Registration Page	Tutor registration screen must contain both shared User fields AND Tutor-only fields (Teaching Languages, Subjects Taught, Upload Qualifications).	All User fields present ✓ plus all Tutor-specific fields ✓. User→Tutor inheritance correctly reflected.	**Pass**
IT-S1-03	User.login() shared by Student and Tutor. One login screen for both roles.	Sign In Page	A single Sign In screen must serve both Student and Tutor without a role selector, reflecting that login() is a User-level method.	Single Sign In screen with no role selector. Both students and tutors use the same screen. Consistent with User.login() as a shared inherited method.	**Pass**
IT-S1-04	User.logout() shared by Student and Tutor.	Side Menu	Logout must appear in the same navigation location for all users, not differentiated by role.	Single 'Log out' option at the bottom of the Side Menu, not role-specific. Consistent with User.logout() as a shared method.	**Pass**
IT-S1-05	Student.searchTutors() creates a SearchQuery. SearchQuery.keyword = search bar input.	Search Page	The search bar represents Student.searchTutors() being called. The search feature must be accessible from the	Search bar present on Search page ✓. Search is accessible from the Side Menu under	**Pass**

			student-facing navigation.	'Search' ✓, which is the student-facing navigation.	
IT-S1-06	Student.applyFilters() updates SearchQuery attributes: subjectFilter, languageFilter, priceMin, priceMax, ratingFilter.	Filter Popup	Each filter control in the popup must map to a SearchQuery attribute: Subject → subjectFilter, Language → languageFilter, Min/Max Price → priceMin/priceMax, Stars → ratingFilter.	Subject dropdown ✓, Language dropdown ✓, Min Price / Max Price ✓, Stars ✓. All SearchQuery filter attributes represented in the Filter Popup.	**Pass**
IT-S1-07	Search returns a list of Tutor objects with Tutor class attributes.	Search Page, Filtered Results Page	Each tutor card must reflect Tutor class attributes: name, subject, overallRating, hourlyRate, bio.	Both result screens show cards with: name ✓, subject ✓, star rating ✓, price/hour ✓, bio ✓. All Tutor attributes represented.	**Pass**

5. User Feedback — Sprint 1

5.1 Collection Method

User feedback for Sprint 1 was collected using the TutoConnect Feedback Form distributed via Google Forms. Participants were asked to walk through the Sprint 1 prototype screens and answer questions covering the Registration screens (Section A), the Login and Session screens (Section B), and the Tutor Search and Filter screens (Section C).

The form was shared with participants after they reviewed the Sprint 1 mockup screens. All responses are recorded directly from the Google Form submissions.

5.2 Sprint 1 Feedback Form Questions

The following questions from the Google Form are relevant to Sprint 1 screens. Responses are recorded in the table below.

Q No.	Section	Question	Response(s) Received
Q1	A — Registration	Were you able to understand the student or tutor registration form without confusion? (Scale 1–5)	
Q2	A — Registration	Were there any fields on the registration form that were confusing or unclear? (Checkboxes)	
Q3	A — Registration	Were the error messages or validation indicators on the registration form clear? (Multiple choice)	
Q4	B — Login & Session	How easy was it to find and use the Sign In screen? (Scale 1–5)	
Q5	B — Login & Session	Did the Forgot Password / OTP recovery flow work as you expected? (Multiple choice)	
Q6	B — Login & Session	Did the Remember Me feature behave as expected? (Multiple choice)	
Q7	C — Tutor Search & Filter	How satisfied were you with the tutor search results? (Scale 1–5)	
Q8	C — Tutor Search & Filter	Were the filter options sufficient for finding the right tutor? (Multiple choice)	
Q9	C — Tutor Search & Filter	Did the tutor cards show enough information to help you make a decision? (Multiple choice)	
Q10	C — Tutor Search & Filter	What improvements would you suggest for the Sprint 1 screens? (Open text)	

Note: The Response(s) Received column is to be completed after Google Form responses are collected from participants.

6. Test Summary — Sprint 1

6.1 Results

Test Type	Total Checks	Pass	Fail	Pass Rate
User Acceptance Testing (UAT)	18	17	1	94.4%
System Testing	12	11	1	91.7%
Integration Testing	7	7	0	100%
SPRINT 1 TOTAL	37	35	2	94.6%

6.2 Failed Tests

Test ID	Description	Impact	Recommended Fix
UAT-S1-08	Sign In screen has no visible error message area for wrong credentials. The invalid login state is not represented in the mockup.	High — users have no feedback when login fails. The SD2 login failure path is not visually covered.	Add a visible error message area (e.g. red text below the Login button) to the Sign In mockup to represent the invalid credentials state.
ST-S1-05	System Testing confirms the same gap: SD2 login failure path requires an error message to be returned to the client, but no corresponding UI element is present on the Sign In screen.	High — the system flow for login failure is architecturally defined in SD2 but has no visual representation.	Same fix as UAT-S1-08. Adding the error area to the Sign In screen resolves both failures.

6.3 Observations

35 out of 37 Sprint 1 prototype checks passed. The two failures both relate to the same missing element on the Sign In screen — the absence of an error message area for invalid credentials. This is a single fix that resolves both UAT-S1-08 and ST-S1-05.

All 7 integration checks passed. The prototype correctly reflects the User→Student and User→Tutor inheritance, the shared login/logout methods, and the full Student→SearchQuery→Tutor filter chain.

User feedback results from the Google Form will be recorded in Section 5 once participant responses have been collected.